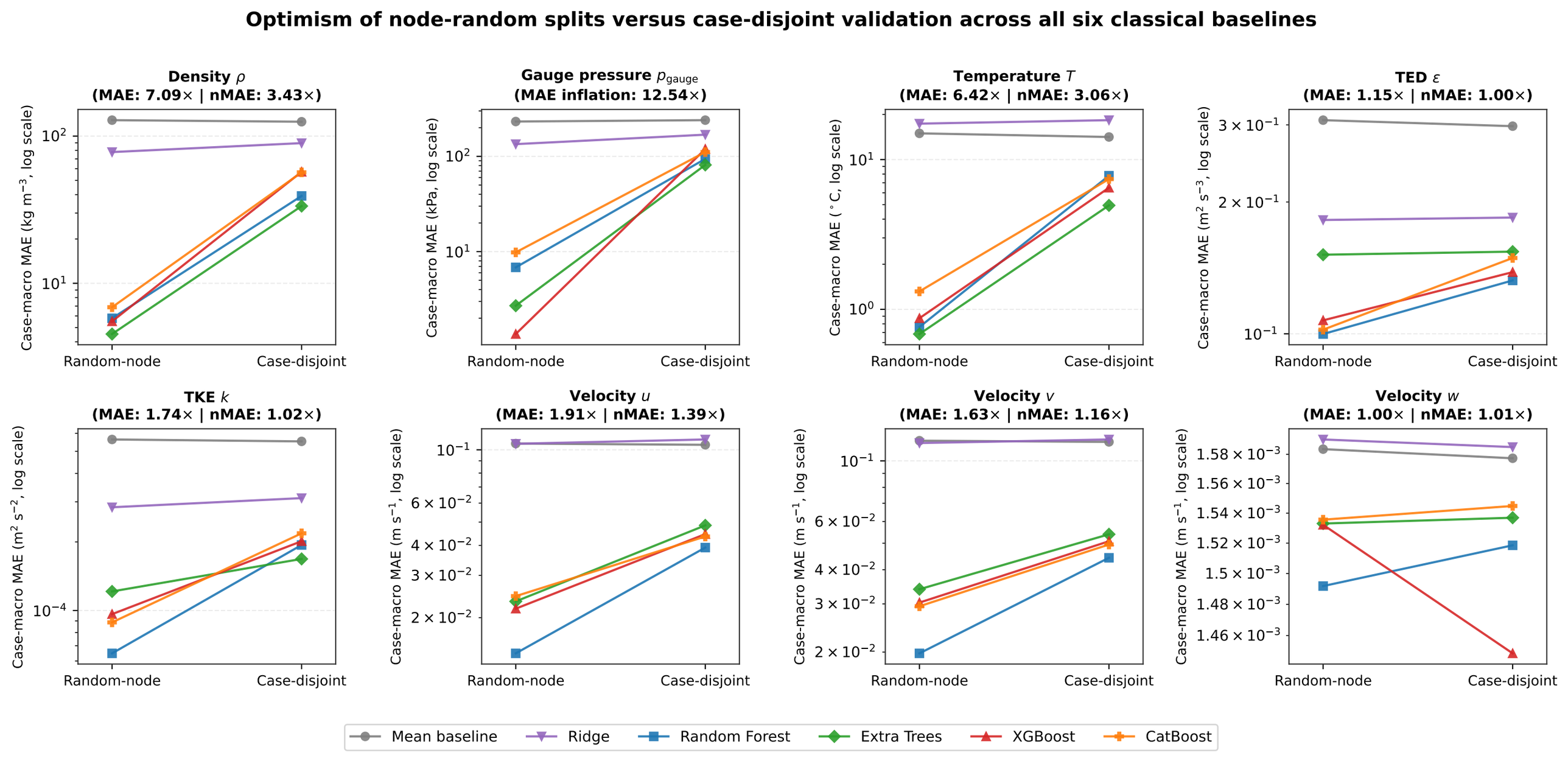
Optimism of node-random splits versus case-disjoint validation across all six classical baselines
Density ρ
(MAE: 7.09× | nMAE: 3.43×)
Case-macro MAE (kg m⁻³, log scale)
Gauge pressure p_gauge
(MAE inflation: 12.54×)
Case-macro MAE (kPa, log scale)
Temperature T
(MAE: 6.42× | nMAE: 3.06×)
Case-macro MAE (°C, log scale)
TED ε
(MAE: 1.15× | nMAE: 1.00×)
Case-macro MAE (m² s⁻³, log scale)
TKE k
(MAE: 1.74× | nMAE: 1.02×)
Case-macro MAE (m² s⁻², log scale)
Velocity u
(MAE: 1.91× | nMAE: 1.39×)
Case-macro MAE (m s⁻¹, log scale)
Velocity v
(MAE: 1.63× | nMAE: 1.16×)
Case-macro MAE (m s⁻¹, log scale)
Velocity w
(MAE: 1.00× | nMAE: 1.01×)
Case-macro MAE (m s⁻¹, log scale)
Random-node
Case-disjoint
Mean baseline
Ridge
Random Forest
Extra Trees
XGBoost
CatBoost